

Q1 Explain briefly with the aid of circuit diagram the principles of operation of a Murray loop test for earth fault. Hence show that

distance of fault from testing end is given by

$$d = \frac{Q}{P+Q} \times 2L \text{ meters.}$$

Q2. Varley loop test is performed to locate an earth fault on a 20 km long cable. The resistance per km of the single conductor is 20 Ω . The loop is completed with a similar healthy conductor. At balance, the variable resistance connected to the faulty conductor is 200 Ω . The fixed resistors have equal halves. Calculate the distance of the fault from the test end.

Q3. Explain briefly the following types of cable faults:

- (i) Ground or earth fault
- (ii) cross or short-circuit fault
- (iii) open circuit faults.

Q4 Explain briefly the following types of transducers:

- (i) Resistive type transducer
- (ii) Inductive transducers operating on Variation of Self inductance.
- (iii) Thermistors
- (iv) Thermocouples

Q5. A resistance wire strain gauge uses a soft iron wire of small diameter. The gauge factor is ± 5.2 . neglect the piezoresistance effect. Calculate the Poisson's ratio.

Q6. Calculate the gauge factor of a strain gauge if a 15mm diameter conductor that is 24 mm long changes length by 1mm and diameter by 0.02mm under a compression force.

Q7. Write a short note on the following types of strain gauges.
(i) Bonded wire strain gauge (ii) Unbonded wire strain gauges.

Q8. Use appropriate circuit diagram explain briefly the following types of recorders:
(i) Strip chart recorder (ii) X-Y-Z Recorder.

Q9. With the aid of circuit diagram explain briefly the principles of operation of the following null type recorders
(i) Bridge type recorder (ii) Linear Variable Differential Transformer (LVDT) recorder

Q10. Use block diagram explain briefly the operating principles of the following digital voltmeters.

(i) Chopper type D.C voltmeters (ii) Successive-Approximation digital voltmeters (DVM).

Q11. Use circuit diagram explain the operation of a Direct-Coupled amplifier D.C voltmeter.

Q12. State the applications of wave analyser.

Q13. With the aid of block diagram explain briefly the principles of operation of a spectrum analyser.

Q14. State the advantages of digital instruments over the analog type of instruments.